Exercise 4.3

1. (i) 15 kg. (ii) ₹ 36 2. (i) Direct proportion (ii) k = 5 (iii) C = 50
3. 90 months 4. 12 minutes 5. ₹ 140 6. 3 7. 300 litres

Challenge Problems

8. ₹ 1155 9. 35 minutes 10. 81 11. 6 days 12. 10 days 13. 5 days

GEOMETRY

Exercise 5.1

1. The pairs of adjacent angle are $\angle ABG$ and $\angle GBC$, $\angle BCF$ and $\angle FCE$, $\angle FCE$ and $\angle ECD$, $\angle ACF$ and $\angle FCE$, $\angle ACF$ and $\angle ECD$
2. 65° 3. 86° 4. (i) 108° (ii) 46° (iii) 30°
5. The other angle is also a right angle. 6. 30°, 120°, 210° 7. 63°
8. Adjacent angles: $\angle PQU$ and $\angle PQT$, $\angle PQT$ and $\angle TQS$, $\angle TQS$ and $\angle SQR$, $\angle SQR$ and $\angle RQU$, $\angle RQU$ and $\angle PQU$ (all possible answers).

   Vertically opposite angles: $\angle PQU$ and $\angle TQR$, $\angle PQT$ and $\angle UQR$.
9. 120° 10. 105° 11. 45°, 135° 12. (i) 125° (ii) 55°

Objective type questions

13. (iii) one common arm, one common vertex, no common interior
14. (iii) linear pair 15. (iv) equal in measure
16. (i) 360° 17. (ii) 80°

Exercise 5.2

1. (i) The angles are exterior angles on the same side of the transversal
   (ii) The angles are alternate exterior angles
   (iii) The angles are corresponding angles
   (iv) The angles are interior angles on the same side of the transversal
   (v) The angles are alternate interior angles
   (vi) The angles are corresponding angles
2. (i) 35° (ii) 65° (iii) 145° (iv) 135° (v)90°
3. (i) 28° (ii) 58° (iii) 123° (iv) 108°
4. (i) 149° (ii) 45° (iii) 101° (iv) 158°
5. (i) 42° (ii) 8° (iii) 2° (iv) 15°
6. (i) 55° (ii) 35°
7. (i) No. Since, interior angles on the same side of the transversal are supplementary.

   (ii) No. Since Corresponding angles are equal.
9. Minimum number of angles is 1. Using the concept of linear pair of angles, we can find one more angle and by the concepts of corresponding angles and alternate angles (interior and exterior) we could find all other angles.

Answers.indd 128 17-12-2019 16:58:53